

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

15th September 2021

**Subject CP1 – Actuarial Practice
(Paper B)**

Time allowed: 3 Hours 30 Minutes (09.30 - 13.00 Hours)

Total Marks: 100

- Q. 1)** A private Indian life insurer which is part of a major financial group sells life insurance products to cater to the demands of the Indian market. The insurer is present for more than a decade in the Indian market and is a substantial player in the life insurance market.

India is impacted by the out-break of the Covid-19 pandemic since last one year which has impacted the various economic and non-economic parameters like the slowdown in the economy due to lockdowns, increase in the morbidity/mortality in the population, fall in the interest rates, volatile equity markets etc.

- i)** What are the key concerns which are required to be addressed as result of the pandemic from perspective of the following stakeholders? (7)

- a) Regulator
- b) With Profit Actuary

The insurance regulatory authority has mandated the insurers to offer simple standardized individual pure term assurance plan with maximum death cover of INR 30 lacs and pre-defined terms and conditions. The regulator has given the direction to the insurers to fulfil the needs of an average customer of having death cover at affordable rates in times of the Covid-19 pandemic.

- ii)** Discuss the challenges which will be faced by the insurer in pricing and offering the standardized plan. (12)

The insurer has significant portfolio of limited premium payment term of traditional non-participating savings plan with guaranteed benefits. Due to recent volatility in the interest rates the insurer is contemplating entering into derivatives which it has not done before, to hedge the interest rate risk on the existing non-participating savings portfolio.

- iii)** Discuss the nature of the interest rate risk faced by the insurer on the traditional non-participating portfolio. (3)

- iv)** What are the risks which company will be exposed due to entering into the interest rate derivatives and discuss how such risks can be mitigated? (8)

The traditional non-participating savings plans also provide death benefit which is fixed at 10 times of the annual premium through the tenure of the policy to comply with the product regulations.

- v)** Discuss whether strengthening of the statutory reserves would be required on account of higher mortality experience due to the pandemic. (6)

The non-life insurer of the same financial group is a key player in the travel insurance segment over the last several years.

- vi)** List down the coverage provided by the international travel insurance policy to meet the needs of the Indian traveller. (5)

- vii)** Discuss the impact of covid-19 on the travel insurance segment for the non-life insurer. (9)

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- Q. 2)** Company A is a proprietary general insurer operating for the past 16 years. The insurer is having significant free capital.

Company B is a proprietary standalone health insurance company operating for the past 10 years. The insurer is having marginal free capital. The public disclosures of Company B indicate that there is a capital infusion in the last two years. The company is expecting capital infusion during this year also to demonstrate solvency to the regulator.

Considering the increasing requirement of capital, Company B is planning to sell its insurance business to Company A.

- i)** Discuss the reasons for increasing requirement of capital infusion by Company B. (7)

- ii)** Set out the reasons for purchase of Company B by Company A. (6)

The Board of Company A asked its Actuary to project the future solvency for the combined entity for next three years to understand the likely capital requirement and profitability on purchase of Company B.

- iii)** Set out the assumptions required for projecting the capital. (6)

- iv)** Discuss the issues that may be encountered by the Actuary while setting assumptions. (6)

- v)** Describe methods of analysis available to assess the range of scenarios generated while projecting the business. (5)

The Board of Directors of Company B has come across various reports indicating that there is increase in health insurance awareness among public due to pandemic. Further, the Board also have come across newspaper reports that the insurance council has made a request to reduce the solvency capital requirement for health insurance business to tide over the strain created due to excess claims experience due to pandemic. Considering all these developments, the Board decided to drop the idea of selling off the insurance business. The Board decided to raise additional capital as well as to manage the existing capital in an efficient way.

- vi)** Discuss about the merits and demerits of request for reduction in solvency capital requirement. (5)

- vii)** Set out the various capital management tools available for Company B. (6)

- viii)** Discuss the ways in which the risks arising from the reasons identified in (i) above can be managed by Company B. (9)

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